

PHASE 2

REQUIREMENT ANALYSIS PHASE

1. Software Requirements

Based on the project structure, the following software components are required:

- **Programming Language:** Python
- **AI/LLM Integration:** Google Gemini Models (Gemini Pro, Gemini Flash)
- **Image Generation:** Local Diffusion models
- **Dependencies Management:** Managed via `requirements.txt` and `requirements-local-diffusion.txt`
- **Environment Configuration:** `.env` file for secure API key and environment variable storage.

3. Functional Requirements

- The system must generate story text utilizing Gemini Pro or Gemini Flash.
- The system must generate images corresponding to the story panels.
- The system must arrange text and images into a cohesive layout.
- The system must export the final layout as a PDF file.